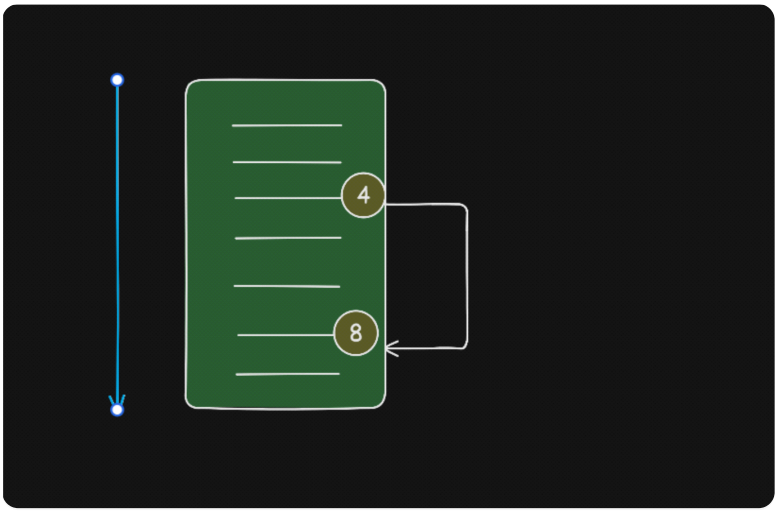
Control Statements

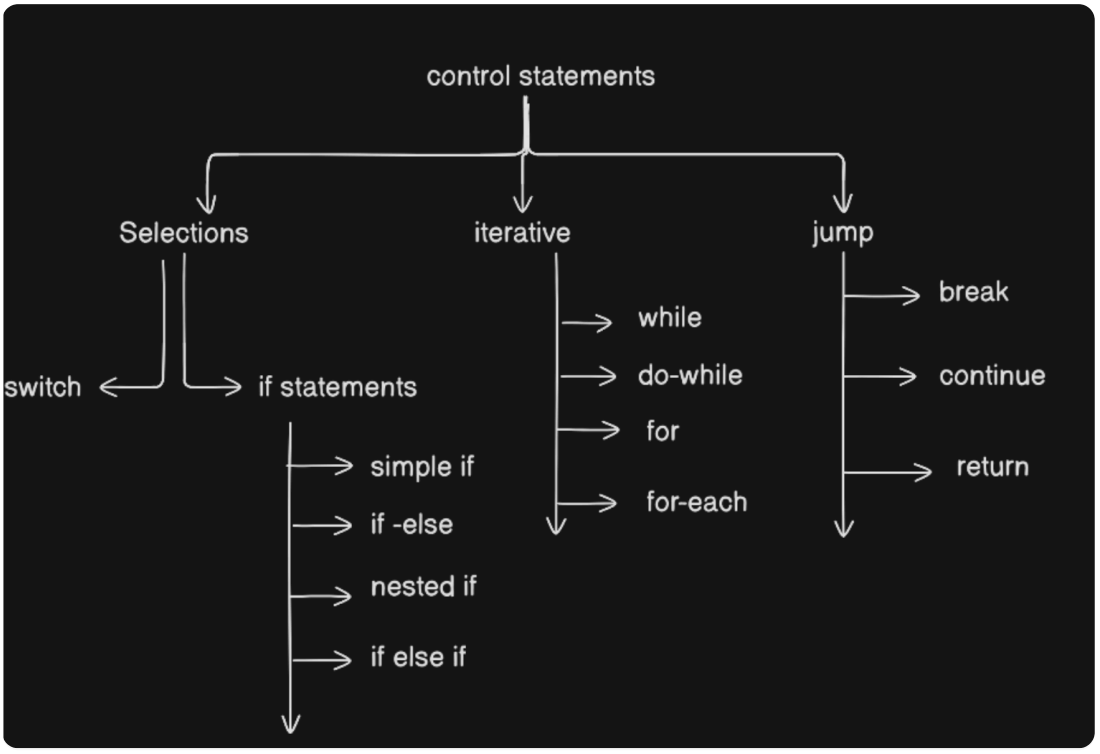
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Till Now, students have learned:

1. Data types
2. Variables
3. Operators

Now we have to learn how java makes decisions:





What are control statements ?

Control statements are used to control the flow of execution .

> **Control statement** = statement that controls which code should execute.

> **Note:** normally java executes code from top to bottom.

```java
class ControlStatements
{
    public static void main(String[] args)
    {
        System.out.println("Line 1");
        System.out.println("Line 2");
        System.out.println("Line 3");
        System.out.println("Line 4");
        System.out.println("Line 5");

    }
}
```

```
D:\Spark6.0>java ControlStatements
Line 1
Line 2
Line 3
Line 4
Line 5

D:\Spark6.0>
```

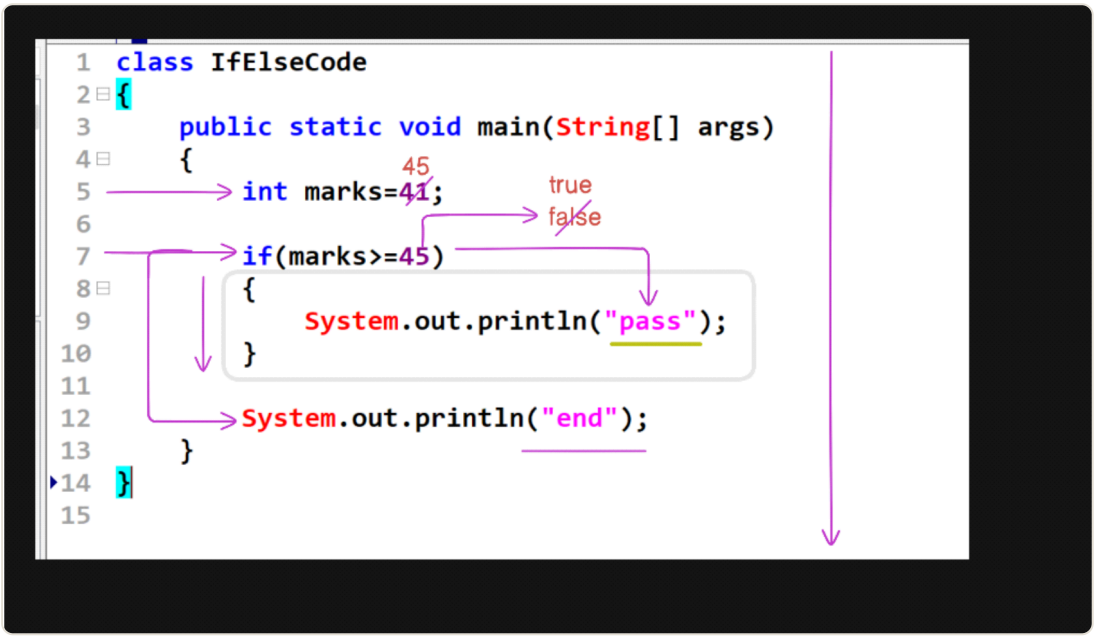
But sometimes we want code to execute only when a condition is true, for that we use **decision making statements**.

What is condition ?

EG:

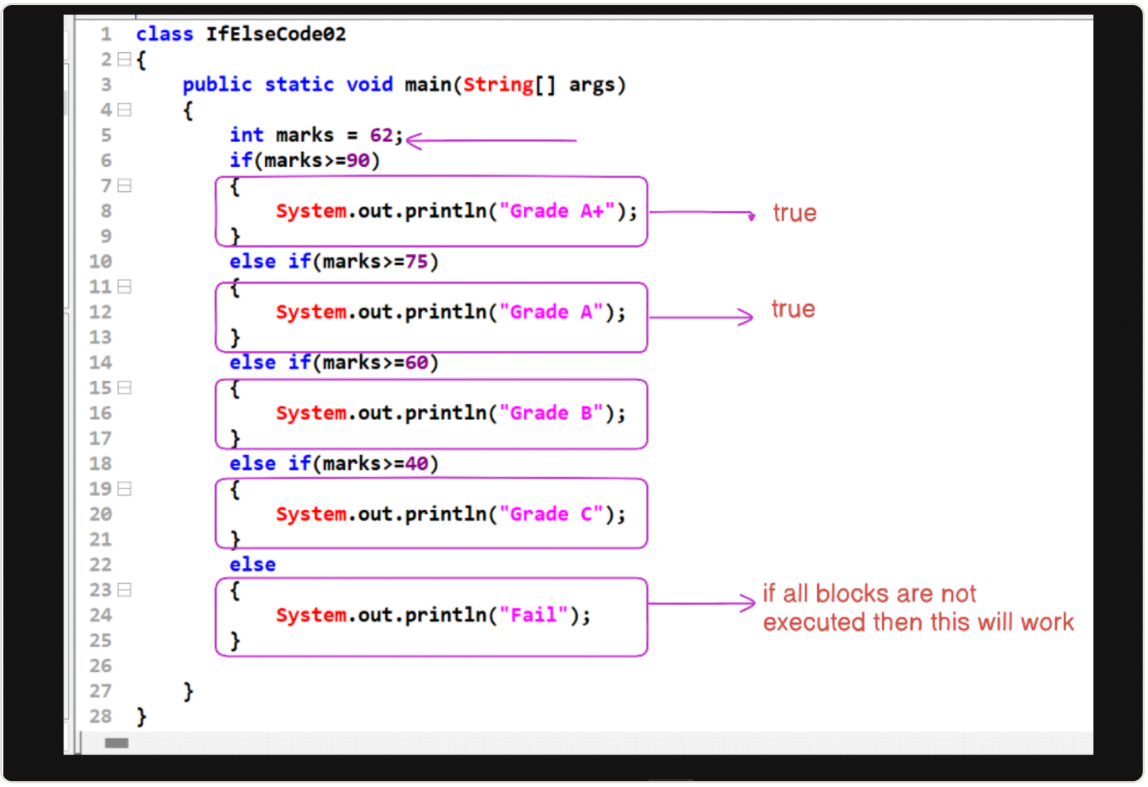
`age>=18`	`marks>=45`	`pass==123`	`number%2==0`

01 How if works:



02 Else-if

We use **else-if** when we have more than two possibilities.



03 Nested If

> **Nested if** means if inside another if

- Use nested if when second condition should be checked only after first condition is true

04 Switch

> **Switch** is used when we compare one value to multiple fixed options

Real life example:

`Day names` `Grade` `Month`

Why we need it ?

`If` `If else` `Else-if` `Switch`

> **Note:**
>
> Conditions help java make decisions
> Loops help java repeat instructions

- ✓ `Print hello world 10 --> ok will write 10 times`
- ✓ `Print hello world 100 --> ok will write 100 times`
- ✗ `Print hello world 1000 ---> ok will write 1000 times`
- ✗ `Print hello world 100000 --> ok will not write`

```
System.out.println("Hello World!");
```

What is loop ?

loop is a terminology which helps us to

1. Write less code
2. Less repetition
3. Easier to write

3 important parts of loop

1. **Initialization**
2. **Condition**
3. **Update**

Types of loops

1. **While loop**
2. **Do-while**
3. **For loop**

01 While loop

```java
        //initialization
        int i=1;

        //condition
        while(i<=10)
        {
            System.out.println("Hello World!");
            i++; //i=i+1;
        }


    }
```

02 Do-while loop:

```
Initialization

do{

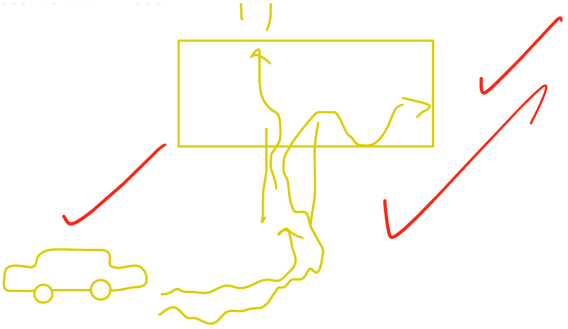

}while();
```

Real world use of do-while

EXAMPLE:

Normal: whey protein --- 100 -- do while
Gold user : whey protein - 1 -- do while

Amazon prime : 20 min ---> 1 ad
Normal user : 5 min -> 1 ad



03 For loop

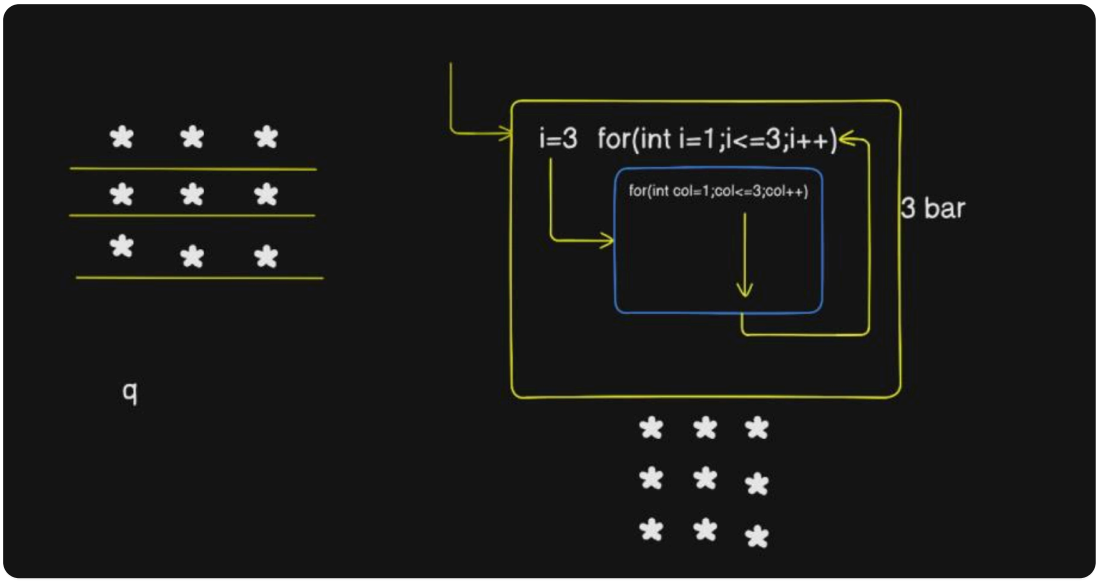
> Use **for loop** when initialization, condition and update are known

Difference between loops

While	Do while	For
Checks condition first	Executes first	Checks condition first
May run zero times	Runs at least once	May run zero times
Good when repetitions are uncertain	Good for menu/input entry, role based activities	Good when count is known
Initialization and condition are separate	Initialization and condition are separate	Initialization and condition are together

Nested Loops

A loop inside another loop:



Ctrl + c

BREAK : 11:03 -----> 11:25